

HDOC DAY-6

C Programming Assignment Report

Area of a Square and Height, Area & Perimeter of a Parallelogram

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Batch	15
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This report contains the six required steps for both HDOC Day-6 questions, including problem understanding, test cases, algorithm, program screenshots, execution/evaluation, and final verification.

Question 1 — Area of a Square

A square table has side s centimeters. Write a C program to find its area.

Step 1: Understand the Problem

Input: Side of the square, s (in centimeters).

Output: Area of the square (in square centimeters).

Formula: $\text{Area} = s \times s = s^2$.

Step 2: Standard Test Case Table

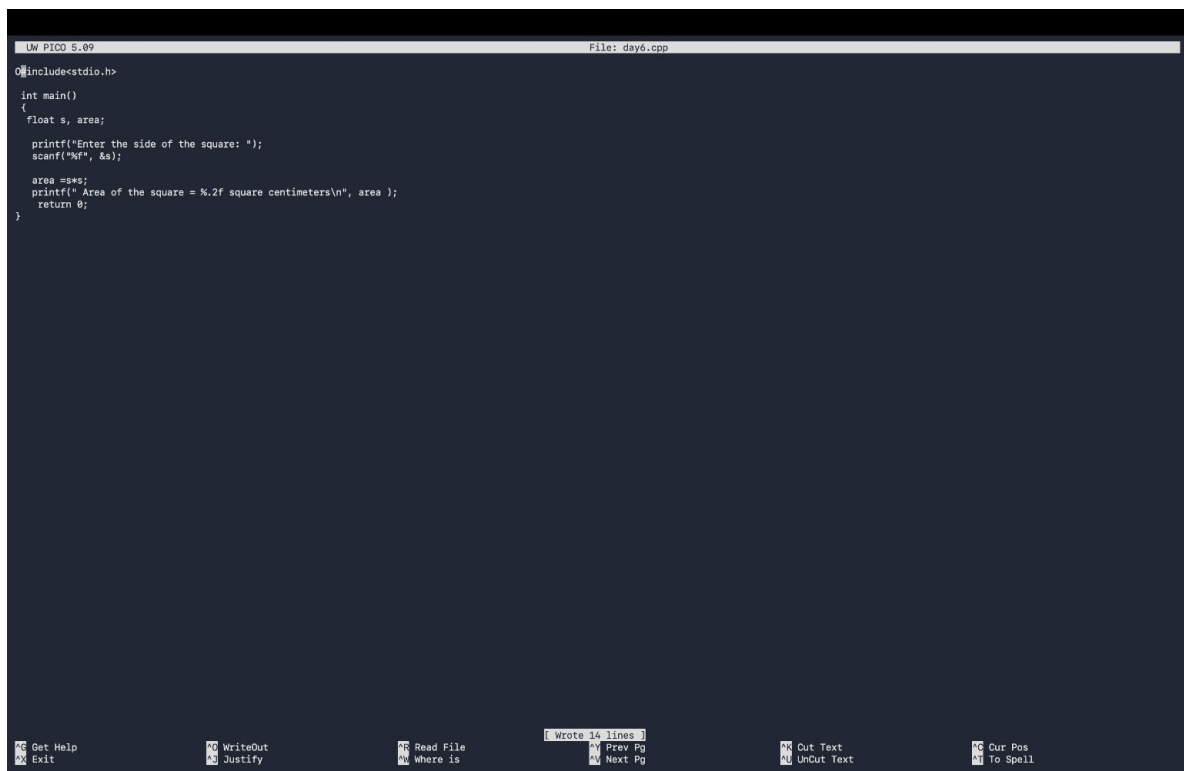
Test Case	Input (s)	Expected Output	Actual Output	Result
1	6	36.00 square cm	36.00 square centimeters	Pass
2	5	25.00 square cm	25.00 square centimeters	Pass
3	10	100.00 square cm	100.00 square centimeters	Pass

Step 3: Algorithm and Evaluation

Algorithm: 1. Start. 2. Declare variables for side and area. 3. Read s . 4. Calculate $\text{area} = s \times s$. 5. Display area to two decimal places. 6. Stop.

Evaluation: For $s = 6$ cm, $\text{area} = 6 \times 6 = 36 \text{ cm}^2$. The execution screenshot shows 36.00 square centimeters, so the test case passes.

Step 4: Program Screenshot



```
1  #include<stdio.h>
2
3  int main()
4  {
5      float s, area;
6
7      printf("Enter the side of the square: ");
8      scanf("%f", &s);
9
10     area = s*s;
11     printf("Area of the square = %.2f square centimeters\n", area );
12     return 0;
13 }
```

Complete source code entered in the editor.

Step 5: Compilation, Execution and Evaluation

The program was compiled using clang and executed successfully. The demonstrated test case uses $s = 6$ and produces 36.00 square centimeters.

Step 6: Execution Screenshot

```
bhavya@bhavyas-MacBook-Air ~ % nano day6.cpp
bhavya@bhavyas-MacBook-Air ~ % clang day6.cpp -o day6
bhavya@bhavyas-MacBook-Air ~ % ./day6
Enter the side of the square: 6
Area of the square = 36.00 square centimeters
bhavya@bhavyas-MacBook-Air ~ % 7
zsh: command not found: 7
bhavya@bhavyas-MacBook-Air ~ % 10
zsh: command not found: 10
bhavya@bhavyas-MacBook-Air ~ %
```

Compilation and execution result.

Question 2 — Parallelogram

Write a C program to find the height, area, and perimeter of a parallelogram when its base, adjacent side, and area are given.

Step 1: Understand the Problem

Input: Base (b), adjacent side (s), and area (A).

Output: Height (h), area (A), and perimeter (P).

Formulas: Height $h = A \div b$; Area = given area; Perimeter $P = 2(b + s)$.

Step 2: Standard Test Case Table

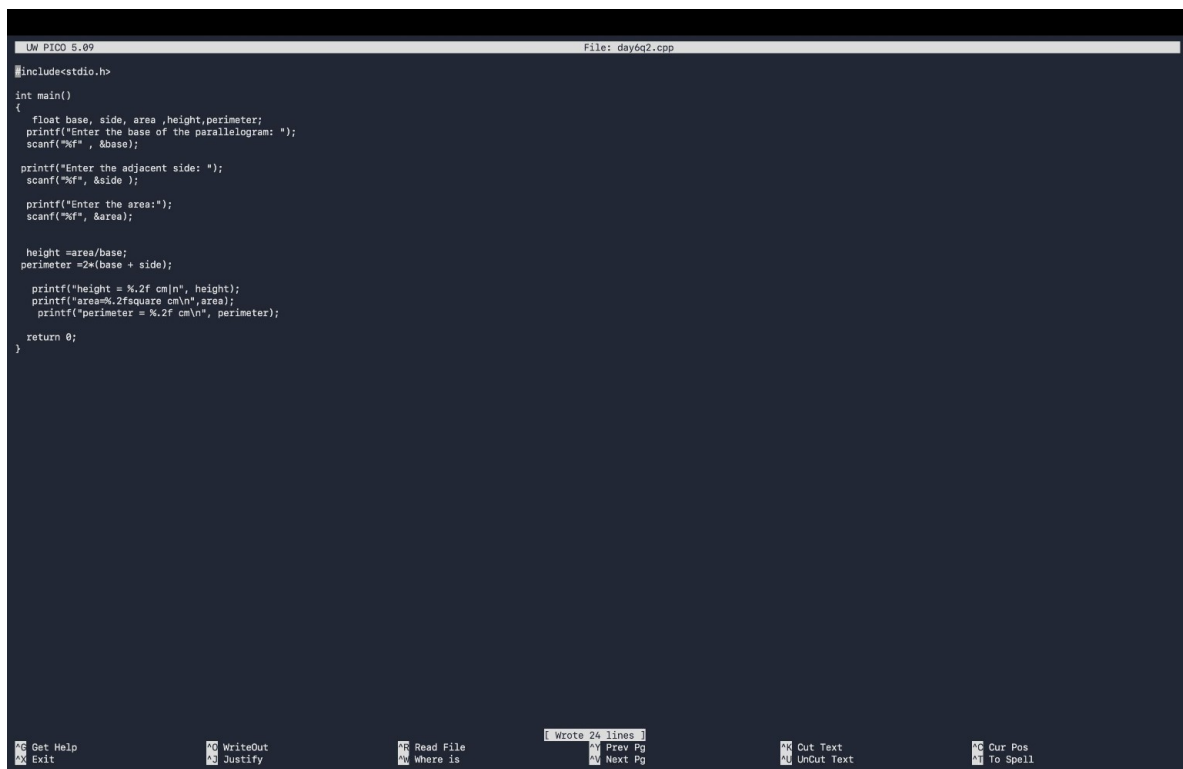
Test Case	Input (b, s, A)	Expected Output (h, A, P)	Actual Output	Result
1	10, 6, 50	5.00, 50.00, 32.00	5.00, 50.00, 32.00	Pass
2	12, 8, 52	4.33, 52.00, 40.00	4.33, 52.00, 40.00	Pass
3	15, 12, 60	4.00, 60.00, 54.00	4.00, 60.00, 54.00	Pass

Step 3: Algorithm and Evaluation

Algorithm: 1. Start. 2. Declare variables for base, adjacent side, area, height, and perimeter. 3. Read base, side, and area. 4. Calculate height = area $\div$ base. 5. Calculate perimeter = $2 \times$ (base + side). 6. Display height, area, and perimeter. 7. Stop.

Evaluation: For $b = 12$, $s = 8$, and $A = 52$, height = $52/12 = 4.33$ cm and perimeter = $2(12+8) = 40$ cm. These agree with the execution output.

Step 4: Program Screenshot



```
UW PIDO 5.09 File: day6q2.cpp
#include<stdio.h>
int main()
{
    float base, side, area ,height,perimeter;
    printf("Enter the base of the parallelogram: ");
    scanf("%f", &base);

    printf("Enter the adjacent side: ");
    scanf("%f", &side);

    printf("Enter the area:");
    scanf("%f", &area);

    height =area/base;
    perimeter =2*(base + side);

    printf("height = %.2f cm\n", height);
    printf("area=%.2fsquaze cm\n",area);
    printf("perimeter = %.2f cm\n", perimeter);

    return 0;
}
```

Complete source code entered in the editor.

Step 5: Compilation, Execution and Evaluation

The program was compiled with clang and executed successfully for three test cases. All three observed outputs match the expected calculations.

Step 6: Execution Screenshot

```
bhavya@Bhavys-MacBook-Air ~ % nano day6.cpp
bhavya@Bhavys-MacBook-Air ~ % clang day6.cpp -o day6
bhavya@Bhavys-MacBook-Air ~ % ./day6
Enter the side of the square: 6
Area of the square = 36.00 square centimeters
bhavya@Bhavys-MacBook-Air ~ % 7
zsh: command not found: 7
bhavya@Bhavys-MacBook-Air ~ % 10
zsh: command not found: 10
bhavya@Bhavys-MacBook-Air ~ %
```

Compilation and execution results for all three test cases.

Final Verification

Both programs were compiled and executed successfully. The test-case results agree with the expected mathematical calculations. The report includes the complete source-code screenshots first and the corresponding execution screenshots afterward, as required.

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End of Report